

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Relevant Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineer Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built backend automation for compliance-sensitive filings with public-site API lookups, draft generation, validation, retries, **session recovery**, and operator checkpoints, reducing manual filing time by **85%**.
- Shipped Dockerized services on **Google Cloud Run** with **GitHub Actions** release workflows, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Integrated a **FastSurfer CNN segmentation** feature for **95 anatomical regions**, wiring **PyTorch/TorchScript** and **ONNX/Tinygrad** inference into a mature image-processing application.
- Built visual ML preprocessing and postprocessing for **NIfTI/MGZ image data**, including voxel conformation, thick-slice datasets, multi-view aggregation, label remapping, LUT mapping, and generated **binary masks**.
- Reduced runtime by **87%** with async/parallel execution while debugging model-output orientation, label correctness, GPU/CPU paths, and visual inspection workflows.

Open Source

Contributions

- Contributed maintainer-reviewed patches across **Invesalious**, **Kuadrant MCP Gateway**, **Kyverno**, **Tiled**, **CRIU**, and **VideoLAN**, spanning visual/model-output correctness, backend reliability, parser robustness, and security-validation paths.

Projects

Hyoka | Python, FastAPI, PostgreSQL, SQLite, SQLAlchemy, OpenTelemetry, Docker

GitHub

- Built an **AI reliability control plane** with trace/event ingestion, **eval/replay workers**, evaluator registries, validation runs, gate decisions, promotions, audit logs, and signed manifests.
- Designed reproducible checks for AI outputs around immutable inputs, **content-addressed artifacts**, project-scoped API keys, latency/cost constraints, and database-backed worker leases.
- Implemented an OpenAI-compatible proxy, Python SDK, Typer CLI, and dashboard/backend paths so AI behavior could be observed, replayed, compared, and gated before promotion.

Swish Support System | Python, FastAPI, React, PostgreSQL, Redis, Langfuse

GitHub

- Built a **hybrid AI + rules assistant** where LLM assessments classify messy user requests while deterministic policy code controls the final action path.
- Grounded responses in **order, item, evidence, customer, and policy context** with PostgreSQL/Redis state, Langfuse traces, confidence thresholds, and regression checks.
- Separated conversational understanding from deterministic rules so AI output stayed inspectable, constrained, and testable instead of prompt-only.

Penny Lane Capital | Python, Agno, Pydantic, SQLite, Vector Memory, JSONL

GitHub

- Built a **Python multi-agent recommendation workflow** with analyst roles, debate, risk review, typed Pydantic state, SQLite checkpoints, and replayable JSONL traces.
- Added **vector memory** for semantically similar decision lessons plus validation harnesses and baselines so recommendations could be inspected and compared.

Technical Skills

Languages: Python, TypeScript, JavaScript, Go

Frameworks: PyTorch, FastAPI, React, Next.js, Node.js, SQLAlchemy

Tools: NumPy, pandas, scikit-learn, ONNX Runtime, PostgreSQL, SQLite, Redis, Docker, GitHub Actions, Langfuse